# Algebra II with Trigonometry Chapter 5.5

**Study Guide (Gemini)**

Saved tutoring response with MathJax rendering. Original PDF file: 4465374_alg-2trig-h-chp-5-5-note-docx-1.pdf

Here is a comprehensive study guide and set of resources based on the provided document about the Law of Sines.

### 1. Short Summary

This document covers Section 5.5, which introduces the **Law of Sines** as a method for solving oblique triangles (triangles without a right angle). It primarily focuses on two triangle configurations: Case 1 (ASA or SAA, where two angles and a side are known) and Case 2 (SSA, where two sides and a non-included angle are known). The document dedicates significant attention to the **Ambiguous Case (SSA)**, explaining how to determine whether the given information results in zero, one, or two possible triangles based on the height of the triangle and the lengths of the given sides.

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### 2. Core Definitions, Theorems, and Formulas

**Definitions:**

- **Oblique Triangle:** A triangle that does not contain a $90^\circ$ (right) angle.
- **Ambiguous Case (SSA):** A triangle situation where two sides and the angle opposite one of those sides are known. It is "ambiguous" because this combination of constraints can yield no triangle, exactly one triangle, or two different triangles.
- **Height of a Triangle ($h$):** The perpendicular distance from a base to the opposite vertex. In an SSA triangle where angle $A$ and sides $a$ and $b$ are given, $h = b \sin A$.

**Theorems & Rules (The Ambiguous Case - SSA):** Given sides $a, b$ and angle $A$:

- **If Angle $A$ is Acute ($A < 90^\circ$):**
- *Two Solutions (Two Triangles):* If $h < a < b$ (notated in the text as $b > a > b \sin A$).
- *One Solution (One Triangle):* If $a \geq b$.
- *No Solution (No Triangle):* If $a < h$ (notated as $a < b \sin A$).
- **If Angle $A$ is Obtuse ($A \geq 90^\circ$):**
- *One Solution (One Triangle):* If $a > b$.
- *No Solution (No Triangle):* If $a \leq b$.

**Formulas:**

- **The Law of Sines (Finding an Angle):** $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$
- **The Law of Sines (Finding a Side):** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

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### 3. Worked Study Guide: Step-by-Step

**Goal:** Solve a triangle using the Law of Sines.

- **Step 1: Identify your known variables.**

Look at the triangle given to you. Do you know two angles and one side (ASA / SAA)? Or do you know two sides and an angle opposite one of them (SSA)?

- *Note:* If you have ASA or SAA, there is only one solution. Jump to Step 3. If you have SSA, proceed to Step 2.
- **Step 2: Check for the Ambiguous Case (SSA only).**

If you are given an SSA triangle, you must test how many triangles are possible. 1. Identify the given angle (let's call it $A$), the side opposite it ($a$), and the other adjacent side ($b$). 2. Calculate the height: $h = b \sin A$. 3. Check if $A$ is acute or obtuse. 4. Compare your values using the rules from Section 2. For example, if $A$ is acute and your $a$ side is shorter than the $b$ side but longer than the height $h$, you must prepare to solve for **two** different triangles!

- **Step 3: Set up the Law of Sines.**

Group your known angle and its opposite side as one fraction (e.g., $\frac{a}{\sin A}$). Set this equal to a fraction containing the variable you want to find.

- *Tip:* Keep the unknown variable in the numerator. Use $\frac{a}{\sin A} = \frac{b}{\sin B}$ if finding a side, or $\frac{\sin A}{a} = \frac{\sin B}{b}$ if finding an angle.
- **Step 4: Solve and find the remaining parts.**

Cross-multiply and isolate the unknown.

- If you are finding an angle, use the inverse sine ($\sin^{-1}$) on your calculator.
- *Important for 2-Solution Cases:* The calculator will only give you the acute angle. To find the second valid angle, subtract the calculator's answer from $180^\circ$ (since supplementary angles have the same sine value).
- Finally, use the fact that the interior angles of a triangle always add up to $180^\circ$ to find the third angle, and use the Law of Sines one last time to find the third side.

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### 4. Practice Problems

*These problems are taken directly from the examples provided in the document.*

**Problem 1 (Based on Example 1a):** In $\triangle ABC$, $\angle A = 52^\circ$, $\angle B = 70^\circ$, and side $c = 26.7$. Find side $a$ (labeled as $x$).

**Problem 2 (Based on Example 3b):** Sketch the triangle, and then solve it using the Law of Sines: $\angle B = 10^\circ$, $\angle C = 100^\circ$, and side $c = 115$. Find $\angle A$, side $a$, and side $b$.

**Problem 3 (Based on Example 4i):** Determine the number of possible triangles and solve for all possible parts: side $a = 28$, side $b = 15$, and $\angle A = 110^\circ$.

**Problem 4 (Based on Example 4ii):** Determine the number of possible triangles: side $a = 20$, side $c = 45$, and $\angle A = 125^\circ$.

**Problem 5 (Based on Example 4iii):** Determine the number of possible triangles (you do not need to fully solve): side $b = 25$, side $c = 30$, and $\angle B = 25^\circ$.

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**Answers to Practice Problems:**

- **Answer 1:** First, find $\angle C = 180^\circ - (52^\circ + 70^\circ) = 58^\circ$. Then use the Law of Sines: $\frac{x}{\sin(52^\circ)} = \frac{26.7}{\sin(58^\circ)}$. Solving for $x$ yields $x \approx 24.8$.
- **Answer 2:**
- $\angle A = 180^\circ - (10^\circ + 100^\circ) = 70^\circ$.
- Find $b$: $\frac{b}{\sin(10^\circ)} = \frac{115}{\sin(100^\circ)} \implies b \approx 20.3$.
- Find $a$: $\frac{a}{\sin(70^\circ)} = \frac{115}{\sin(100^\circ)} \implies a \approx 109.7$.
- **Answer 3:** Because $\angle A$ is obtuse and $a > b$ ($28 > 15$), there is exactly **one** triangle.
- Find $\angle B$: $\frac{\sin B}{15} = \frac{\sin(110^\circ)}{28} \implies \sin B \approx 0.5034 \implies \angle B \approx 30.2^\circ$.
- Find $\angle C$: $180^\circ - (110^\circ + 30.2^\circ) \implies \angle C \approx 39.8^\circ$.
- Find $c$: $\frac{c}{\sin(39.8^\circ)} = \frac{28}{\sin(110^\circ)} \implies c \approx 19.1$.
- **Answer 4:** $\angle A$ is obtuse. For an obtuse triangle, the side opposite the given angle must be the longest side. Here, $a \leq c$ ($20 \leq 45$), so there is **no solution (0 triangles)**.
- **Answer 5:** $\angle B$ is acute. Calculate the height $h = c \sin B = 30 \sin(25^\circ) \approx 12.68$. Since $h < b < c$ ($12.68 < 25 < 30$), there are **two solutions (2 triangles)**.

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### 5. Assumptions and Ambiguities Resolved

As a careful tutor, I noted a few visual and typographical ambiguities in the provided document and resolved them to ensure mathematical accuracy: 1. **Typographical errors in the formulas:** In the "Case I: Acute Angle" section, the text labels the condition for two triangles as "$b > a > b \sin A$". A few lines down, the OCR captures rules for one and zero triangles with contradictory letters. I corrected this in the definitions section by relying on standard mathematical rules: If the given angle is $A$, the condition for two solutions is indeed $h < a < b$ (or $b \sin A < a < b$). The condition for one solution is $a \geq b$, and the condition for no solution is $a < h$ (or $a < b \sin A$). 2. **Image clarity in Example 1a:** The exact placement of the $70^\circ$ angle and side $c = 26.7$ is slightly ambiguous due to resolution. I assumed standard triangle vertex notation (angle $B = 70^\circ$, side $c = 26.7$, and side $a = x$) to formulate a valid, solvable geometry problem. 3. **Example labels (Example 2b):** Visually, Example 2b looks like a Side-Angle-Side (SAS) setup based on the drawing, but solving it with the Law of Sines requires an SSA alignment. By following standard vertex naming conventions (Side $AC = b$, Side $BC = a$), it successfully reads as an SSA problem ( $A = 80^\circ, b = 3.4, a = 6.5$), which allows standard Law of Sines application.